Apoorv Raj

+91 8789818605 | apoorvraj1208@gmail.com | LinkedIn | Github

PROFESSIONAL SUMMARY

Artificial Intelligence & Machine Learning undergraduate with experience in building backend-driven applications and ML-integrated systems. Strong foundation in REST API design, authentication workflows, database architecture, and data-driven decision logic.

EDUCATION

BMS Institute of Technology and Management, **Bengaluru** — 2023 - 2027
Bachelor of Engineering (Artificial Intelligence & Machine Learning**)** — Bengaluru, India
CGPA: **9.23/10**

CLASS XII – D.A.V. Public School Gandhinagar, Ranchi (CBSE) — 2021 - 2022
Percentage: **92.6%** — Ranchi, India

CLASS X – D.A.V. Public School Gandhinagar, Ranchi (CBSE) — 2019 - 2020
Percentage: **92.6%** — Ranchi, India

EXPERIENCE

Web Development Intern — Central Coalfields Limited (CCL) *02 August 2025 – 30 August 2025 (4 weeks)*

- Developed CONVOA, a full-stack Q&A platform using PHP and MySQL.
- Implemented secure authentication and structured CRUD operations.
- Designed relational database schema and optimized backend SQL queries.
- Built responsive UI ensuring mobile compatibility.

Full Stack Web Development Intern — Future Interns *27 November 2025 – 30 December 2025 (4 weeks)*

- Designed and implemented full-stack web applications with focus on backend logic and authentication workflows.
- Structured relational databases and optimized SQL queries for performance and reliability.
- Applied modular architecture principles separating routing, business logic, and presentation layers.

PROJECTS

SARTHI — Crowdsourced Public Safety App
Python, Flask, scikit-learn, Firebase Auth, JavaScript

- Built backend using Flask blueprints with secure Firebase-based authentication.
- Trained and integrated a scikit-learn model for safety score prediction using joblib serialization.
- Implemented geospatial logic including grid-cell encoding and time-decay scoring.
- Developed REST APIs for audit submission, heatmap generation, and safe-route computation.
- Integrated frontend communication through authenticated fetch requests and modular JavaScript components.

ShoppR — E-Commerce Platform
PHP, MySQL, Bootstrap, JavaScript

- Designed normalized relational schema for users, products, and orders.
- Implemented secure authentication using prepared statements and password hashing.
- Developed backend modules for cart management, order processing, and role-based access control.
- Built admin dashboard for product management and order tracking.

CONVOA — Q&A Platform
PHP, MySQL

- Built full-stack Q&A platform with secure authentication and structured CRUD operations.
- Designed scalable MySQL schema for user-generated content.
- Optimized backend queries to improve data retrieval efficiency.

TECHNICAL SKILLS

Programming: Python, Java, C++, PHP
Backend & APIs: Flask, REST API Design, Firebase Auth, Session Management
AI & Data: scikit-learn, pandas, numpy
Databases: MySQL, MongoDB
Core Concepts: Data Structures & Algorithms, DBMS, Operating Systems (Basics), System Design (Basics)
Tools: Git, GitHub, VS Code, XAMPP

CERTIFICATIONS

- **Machine Learning for Engineering Sciences Applications – NPTEL (2024)**